Expressions and equations

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 8×3

2. 8×7

3. 8×4

4. 8×9

5. A rectangle has area $\frac{4}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

6. Find $\frac{2}{3} + \frac{1}{4}$.

7. 2×4

8. Find $2n + 7$ when $n = 5$.

9. Expand $2(x + 7)$.

10. Find 5^2. Write the repeated factors.

11. Find $4n + 3$ when $n = 2$.

12. Each pack has b beads. Write the total in 3 packs. Find it when $b = 4$.

13. Lena wrote $5(z + 2) = 5z + 2$. Find the mistake and repair it.

14. A packet contains 5 cards. Write the number in p packets plus 4 loose cards; evaluate at $p = 4$.

15. Evaluate $4t + 5$ when $t = 4$. Show the multiplication first.

16. Evaluate $4^2 + 5$. Explain what the exponent means.

17. Expand $4(x + 5)$ and check at $x = 4$.

18. Simplify $5y + 4y + y$. Evaluate at $y = 2$.

CHALLENGE / Explain your plan and check the result

C1. A museum sells family packs with 2 adult tickets and 3 child tickets. Adult tickets cost a dollars; child tickets cost $4. Write the cost of 4 packs. If $a = $9, will $125 cover the order? Explain.

C2. A rectangular garden is 5 m wide and $n + 2$ m long. A path covers 10 square m inside it. Write the planted area. Find it when $n = 6$, and explain why $5n$ is equivalent to your expression.